Multimedia Signal Processing - 2nd Module

January, 17 2023

1. **Question (7 pts)**
   Consider a continuous source X that emits symbols following a probability distribution function (p.d.f.) $p_X(x)$.
   (a) Consider the following series of symbols emitted by X

   $$x = \{0.1,\ -0.9,\ 0.7,\ -0.8,\ -0.7,\ 0.2,\ -0.7,\ -0.3,\ 0.8,\ -0.2,\ 1.5\}.$$

   Design a Lloyd-Max quantizer tailored to these symbols using $M = 3$ bins and the thresholds $\{-\infty, -0.5, 0.5, +\infty\}$ as initialization. Stop after convergence is reached or at most three complete iterations.
   (b) Apply the obtained quantizer to x thus obtaining $\tilde{x}$.
   (c) Compute the entropy of the source representing $\tilde{x}$ estimating the alphabet and the probability mass function (p.m.f.) directly from $\tilde{x}$.
   (d) Derive an Huffman code to transmit $\tilde{x}$.
   (e) Considering the obtained Huffman code, compute the average code length explaining its meaning.
   (f) Represent using Huffman code a string of three symbols of your choice.

2. **Question (3 pts)** Consider lossless coding for two discrete sources X and Y.
   (a) Report the formal definition of entropy $H(X)$, explaining its meaning and all terms defined in the equation.
   (b) Report the theoretical bounds that can be achieved by the entropy, reporting in which situations the bounds are reached.
   (c) Report a possible way to compute the joint entropy $H(X, Y)$ starting from the knowledge of marginal entropy $H(X)$ and $H(Y)$, conditional entropy and / or mutual information.

3. **Question (5 pts)** Consider the process of encoding a video signal
   (a) Report the block diagram of the typical encoder used in a hybrid DPCM/DCT coding scheme clearly explaining which is the input and output of each block.
   (b) Explain what is the displaced frame difference.
   (c) Explain how it is possible to implement sub-pixel motion estimation.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Consider two independent discrete sources X and Y with entropy $H(X) = 1.5$ and $H(Y) = 1.5$. The joint entropy must be $H(X, Y) = 1.5$.
   - [T] [F] Consider a speech coding scheme based on the use of a DPCM. Both a long-term and a short-term predictor can be used simultaneously and are not alternative choices.
   - [T] [F] Shannon Code and Huffman Code have the same average code length in case of dyadic sources.
   - [T] [F] Audio coding schemes based on perceptual coding typically use the same filterbank for both the perceptual model and the further step of quantization and coding.
   - [T] [F] JPEG coding standard may apply sub-sampling to the red and blue components of an image with respect to the green one, as red and blue are less important for human perception.
   - [T] [F] In open-loop DPCMs, the amount of distortion (in terms of MSE) measured between the input signal and the decoded signal is the same amount of distortion measured between the input of the quantizer and its output.
   - [T] [F] The coding gain of a transform coding scheme can be computed in closed form (i.e., can be defined with a known equation) under the assumption that the optimal bit-allocation strategy can be successfully applied.
   - [T] [F] Perceptual audio coding schemes assume that audio components below the absolute threshold of hearing can be discarded.

5. **Question (Matlab - 10pts)**

   Consider you have an image file called `peppers.png` in your Matlab working directory. This is a RGB image with resolution 512×384.

   Data loading

   (a) Load the image and display it.

   (b) Considering the colorspace, the type of the variable, and the image resolution, compute how much memory is needed to store the uncompressed image.

   (c) Extract one color component from the image, clearly stating which component you are considering.

   DPCM coding Considering the color component you extracted from the image, implement a close-loop DPCM-based coding scheme. Consider the image as a monodimensional vector, use the previous pixel as predictor, and assume that the first pixel is known by the decoder.

   (d) Turn the two dimensional image into a monodimensional vector.

   (e) Consider that you want a quantizer with 8 possible levels, define a quantizer that is reasonably adapted to the residual dynamic (hint: use an open-loop estimate of the residual to tune your quantizer).

   (f) Encode and decode the vector using the defined DPCM and quantizer.

   (g) Turn the decoded vector back into a matrix and display the decoded image.

   (h) Compute the introduced MSE.

   (i) Compute the entropy of the encoded data.